

DATA TRANSFORMATION

Objective:

The goal of this phase was to take the raw/generated data and reshape it into a clean, analysis-ready format. Transformation taken involved:

- Ensuring consistent structure
- Renaming columns for clarity
- Reshaping or joining data as needed

Tools & Environment:

1. IDE: Visual Studio Code (VS Code)
2. Language: Python
3. Libraries Used: pandas, numpy, datetime

Transformation Activity:

1. Check data types and apply conversions as needed
2. Rename columns using lower case formatting.
3. Replace inconsistent values to ensure row-level consistency.
4. Handle null values by filling them based on agreed upon rules.

Limitations:

1. The data is synthetic and does not reflect actual values related to company performance.
2. There were minimal transformations applied in this project, as the data was intentionally generated in good condition for project purposes.

Transformation Workflow:

1. Load Raw Data:

- Loaded the raw csv files generated from the data creation phase

```
import pandas as pd
import numpy as np

customers = pd.read_csv('customers.csv')
travel_package = pd.read_csv('travel_package.csv')
travel_guide = pd.read_csv('travel_guide.csv')
sales = pd.read_csv('sales.csv')
feedback = pd.read_csv('feedback.csv')
refund = pd.read_csv('refund.csv')
```

2. Customers Table:

- Clean 'email' column by replacing values
- Fill the null values with 'not_provided'

```
# 1. customers table
# clean up the email column
customers['email'] = customers['email'].str.replace(';', '').str.replace('.,')
customers.isna().sum()

# fill na email with 'not provided'
customers['email'] = customers['email'].fillna('not_provided')
```

Sample Final Output:

	customer_id	customer_name	gender	age	email	customer_type
0	MY1000	Zelaya Purwanti	male	36	zelaya.purwanti@outlook.com	new
1	MY1001	Dr. Mila Melani, M.Farm	female	29	dr.mila.melani.m.farm@hotmail.com	regular
2	MY1002	Cengkal Mulyani	female	43	cengkal.mulyani@yahoo.com	regular
3	MY1003	Siti Susanti	male	41	siti.susanti@yahoo.com	new
4	MY1004	R.M. Bakiono Hassanah, S.Pd	female	37	r.m.bakiono.hassanah.s.pd@yahoo.com	new

3. Feedback Table:

- feedback_date column is converted from string to date data type.

```
# 2. feedback table

feedback.dtypes

feedback['feedback_date'] = pd.to_datetime(feedback['feedback_date'])
feedback.head()
```

Sample Final Output

	feedback_id	package_id	booking_id	satisfaction_score	feedback_date	comments	rating_hotel	rating_guide	rating_itinerary	would_recommend
0	FB001	PC25	BK752	8	2023-07-11	Highly recommend this package to others.	3	5	5	Yes
1	FB002	PC18	BK617	10	2023-11-27	The hotel and guide exceeded our expectations!	5	4	5	Yes
2	FB003	PC31	BK677	10	2023-06-34	Highly recommend this package to others.	5	4	5	Yes
3	FB004	PC21	BK481	10	2024-02-29	The hotel and guide exceeded our expectations!	5	4	5	Yes
4	FB005	PC42	BK1109	8	2024-01-31	Beautiful destinations and a relaxing itinerary.	3	4	3	Yes

4. Refund Table:

- refund_date column is converted from string to date data type.

```
refund['refund_date'] = pd.to_datetime(refund['refund_date'])
refund.dtypes
```

Sample Final Output:

	refund_id	booking_id	customer_id	refund_date	refund_amount	refund_percentage	reason	refund_status
0	RF001	BK814	MY1522	2023-05-18	2245	9.95	Unforeseen circumstances	Pending
1	RF002	BK369	MY1592	2023-11-22	809	22.26	Health emergencies	Rejected
2	RF003	BK770	MY1452	2023-04-13	485	17.40	Other	Pending
3	RF004	BK704	MY1248	2023-09-23	1937	49.73	Natural disasters	Pending
4	RF005	BK1137	MY2163	2023-10-07	453	11.78	Unforeseen circumstances	Pending

5. Travel Guide Table:

- Clean 'role' column by replacing values

```
# 4. travel_guide table
```

```
travel_guide['role'] = travel_guide['role'].str.replace('_', ' ')
travel_guide.head()
```

Sample Final Output:

	staff_id	staff_name	role	working_status	comission_rate
0	0101	Nasrul	main guide	contract	0.08
1	0102	Khair	assistant guide	contract	0.11
2	0103	Intan	assistant guide	freelance	0.12
3	0104	Amira	main guide	freelance	0.12
4	0105	Nuha	main guide	contract	0.08

6. Travel Package Table:

- No transformation done.

Sample Final Output:

	package_id	package_name	package_type	duration_days	price_per_pax	category	destination
0	PC10	Manila	solo	6	2387	adventure	Manila
1	PC11	Krabi	group	4	2256	eco	Krabi
2	PC12	Taman Negara	solo	2	1018	adventure	Taman Negara
3	PC13	Putrajaya	family	3	763	adventure	Putrajaya
4	PC14	Cebu	family	5	1784	cultural	Cebu

7. Sales Table:

- Clean 'payment_method' column by replacing values

```
sales['payment_method'] = sales['payment_method'].str.replace('_', ' ')
sales.head()
```

Sample Final Output:

	booking_id	customer_id	package_id	guide_id	booking_date	travel_date	num_of_pax	discount_promotion	channel	payment_status	payment_method
0	BK100	MY1525	PC37	0105	2023-06-17	2023-07-10	2	645	online	paid	credit card
1	BK101	MY1108	PC27	0109	2023-08-13	2023-09-11	4	0	offline	paid	online transfer
2	BK102	MY2042	PC18	0106	2023-06-26	2023-08-24	17	1895	online	paid	cash
3	BK103	MY1263	PC26	0101	2023-12-27	2024-01-18	4	230	agent	pending	credit card
4	BK104	MY1229	PC32	0106	2023-04-23	2023-06-19	12	0	agent	paid	cash

8. Saving Outputs: "path: End To End Project/PPNJ/data creation"

```
# Saving all csv files
customers.to_csv(r"C:\Users\User\Desktop\Data Analyst\End To End
Project\PPNJ\data_creation\dim_customer.csv", index=False)

travel_package.to_csv(r"C:\Users\User\Desktop\Data Analyst\End To End
Project\PPNJ\data_creation\dim_travel_package.csv", index=False)

sales.to_csv(r"C:\Users\User\Desktop\Data Analyst\End To End Project\PPNJ\data_creation\fact_sales.csv",
index=False)
feedback.to_csv(r"C:\Users\User\Desktop\Data Analyst\End To End
Project\PPNJ\data_creation\dim_feedback.csv", index=False)

travel_guide.to_csv(r"C:\Users\User\Desktop\Data Analyst\End To End
Project\PPNJ\data_creation\dim_travel_guide.csv", index=False)

refund.to_csv(r"C:\Users\User\Desktop\Data Analyst\End To End Project\PPNJ\data_creation\dim_refund.csv",
index=False)
```

Summary of Transformations:

https://github.com/nasrulkhair/Full-Project/blob/main/PPNJ/data_creation/data.ipynb

- All transformations were done programmatically using pandas to ensure reproducibility.
- Output is clean, schema-consistent, and ready for further modelling

No.	Table	Column	Transformation
1.	customers	email	fill null values with 'not provided'
2.	customers	email	replacement: , → "/" : → "
3.	feedback	feedback_date	convert data type from STRING to DATE
4.	refund	refund_date	convert data type from STRING to DATE
5.	travel_guide	role	'_' → ''
6.	sales	payment_method	replace '_' → ''

